

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 9

~ BookStream #002.03

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hello this video is going to be an overview and introduction to chapter 9 on model based analysis in par Missoula and friston octave inference 2022 textbook so I'll read the opening quote and then Ali feel free to give a first thought or overview on the chapter just because we have the best Hammer does not mean that every problem is an ale Barack Obama so what is your overview on chapter nine okay so chapter nine as the title suggests is about doing uh is about introducing an approach to data analysis specifically uh model based data analysis because as we know there are multiple different approaches to do any kind of analysis and data uh so there are approaches that are basically model free or model without any specific models and we just do some uh some techniques on data wrangling and and the data directly dated directly but in this chapter um there is this approach that that is inspired by active inference um framework that's basically uh is uh is about how to uh organize or how to interpret uh basically the data we have at hand using active inference framework so this kind of model based analysis can be used on many different scenarios involving uh Empirical research in which uh some uh some raw data has been gathered and we need to make sense of those data in terms of uh what active inference models can be applied or can be appropriate to organize those data and what kind of inference can we get by developing a kind of active input model and basically treating the data as the input to the model and then to see what kind of imprints we can get using those data G great thank you they write here in this chapter we focus on the ways in which active inference can be applied in understanding empirical data and the point you ended with is exactly the point that I like to highlight which is that up to this point in the textbook the examples have all been proposed basically from scratch like the rat and the teammates or the frog jumping out of the hand and so first there was the conceptualization of a generative model here in this textbook to kind of demonstrate some feature of active inference framework and then data could be generated from the model whereas in chapter nine we're

thinking a lot more about the back and forth between the generative model and the data including situations where we begin with a given data structure and then construct a generative model to analyze that in different ways that we'll cover okay section 9.2 meta Bayesian methods what would you say about metabayesian methods all right so uh this particular figure on this page figure 9.1 uh tends to be one of the figures uh that we um come time and time again uh when we refer to kind of meta modeling go back to the meta modeling aspect of uh active inference so basically um in this kind of view of active inference uh it is I mean it can be used to model a kind of nested hierarchical models of different sorts so basically it's a kind of self-referential way of modeling a situation of interest in other words instead of just uh modeling a system of Interest we can also apply the same mathematical technology on the same theoretical framework to model the The Observer of that phenomena or let's say the scientist or the agent that intends to study that phenomena in other words it's a kind of uh it's a way to kind of formalize [Music] a specific way of doing a scientific research um in terms of doing uh these this kind of self-reflective or recursional way of meta modeling so in this section of chapter nine uh there are some discussions and also references to many different papers that use this kind of process as this self-referential process of metabayesian modeling to model many many different scenarios so as we see in figure one 9.1 if we take the subjective model as the kind of graph based graph Bayesian graph that we've seen before in previous chapters and then take that exact same model and put it in as the input to the system we can on the outer level of this nested hierarchy look at this as the subject of interest and then apply the exact same mathematical formalism to model this phenomena as well great inside of the box here is the system of Interest this is the subject of the experiment this is the rat and the Maze whose generative model describes observations and hidden States through time and policy conditioned upon the parameterization of the model and then outside of that cognitive model which may well be a map of its surrounding we have the experimenters room with the stimuli that is being passed in it's a policy selection from the experimenter but it's an observation coming in from the perspective of the subject and then the actions coming out of the generative model at the heart of this image is taken up to be objective behavioral data by the experimenter and there's a really interesting point that they make about two reasons why you might want to fit a computational model to observe Behavior the first is to estimate parameters of interest that explain a given particular things Behavior or differentiate groups or subtypes of similar things that's computational phenotyping so that is conditioning upon a structural model looking to differentiate individuals or an individual through time or across contexts and

conversely you might fix upon one individual or a group of individuals and be interested to test different hypotheses about the underlying cognitive structures section 9.3 variational oh yeah please good yeah and also one other point that might be worth mentioning here is um this recent development in Bayesian mechanics literature and specifically uh the proof that the cmap a way of uh looking at the phenomena uh is the exact Dual of the fep uh way of modeling the same the same state space so uh in this way cmap and fep Duality can be mapped onto this self-referential way of metabasian metamasian modeling because on the one hand if we are if we look at the environment or the system of interest from the Viewpoint of the system we were basically dealing with fep but if we flip that logic around we're dealing with cmap or strength maximum entropy principle so yeah that was also I think relevant to metabasian or phenotypic computation as well cool anything to say on 9.3 variational LaPlace and 9.4 parametric empirical days okay so these two sections are pretty technical uh but there's this recent paper I think it was published uh a couple of weeks ago by first at all which provides a tutorial on variation on LaPlace variational techniques uh which I think can greatly help to understand uh or to unpack some of the dense technicalities of this section in this particular section 9.3 so I highly recommend looking at that paper before or alongside reading this section as well uh yeah exactly so yes those are two technical sections as you brought up exactly 9.5 the instructions for model base analysis here they're going to outline some steps to analyze empirical behavioral data Choice data using active inference step one collect behavioral data Step 2 formulate a pomdp partially observable Markov decision process step 3 specify a likelihood function step four specify prior beliefs step 5 solve for posterior probability and model evidence and step six perform group level analysis this is summarized in figure 9.2 what would you say about this process uh okay so I believe this is a kind of common process used in SPM 12 so um well it it helps to look at this process in terms of the recipe provided in chapters chapter 6 and also used extensive mean and modeling the behavior modeling the cases in chapters seven and eight but basically uh those I mean this kind of routine again is not something I believe that needs to be followed precisely in every detail but it can be modified based on the scenario we're interested in so at least that's my understanding of these routines or recipes or whatever we want to call them so but it provides a helpful starting point uh which can help to at least begin to sketch some uh some routes through our modeling uh modeling task without losing uh the forest for the trees yep I would describe this figure as connecting the empirical structure of the data so in this case it'd be the location of the animal in the tmas Through Time connecting the empirical data structure to a generative

model that graciously receives that data without any false positives or false negatives that's where we get the kind of nice fitting design and the category Theory plus specification of some of the accessory Machinery that's required for the Bayesian statistical model and then stage six here is like a figure that you'd see in a paper if there were some parameter in the pomdp that described the mouse's skittishness or its boldness and that parameter was going to be identified and then to go from the parameter to generated data that would be the kind of generative direction or in the recognition Direction with the inverted model you could go from empirical data to identification of parameters and so maybe this graph in six represents 10 mice per replicate and the parameterization of each of their parameters related to some personality trait or performance on a task and then these bar charts could be straightforwardly compared using an anova or t-test and that would basically be the paper we designed the experiment we designed the generative model we did statistics here's the statistical outcomes that we identified in terms of here group differences from these five groups section 9.6 provides examples of generative models they provide in figure 9.3 in overview these are two different generative models of icicade motor Behavior epistemically driven and there's more that could be said but this highlights to me that there's always a plurality of generative models that can be drawn even for the same behavior and this point is made especially clear because on the left the experiment of atoms at all is performs with A continuous time continuous space model whereas the Mirza on the right was in a categorical space and discrete time anything else to say about this setting other than that it was used to cite these papers and give examples of what it looks like to connect a generative model here of isocades and vision to empirical data so I think that's pretty much all there is to it but uh again there are some subtle differences uh between those two examples in terms of how those generative models um are conceived so uh it's helpful to see uh two related examples um I mean they're both about studying the eye movement but different aspects of it so these kinds of subtleties can rise in a real real world situations when we want to when we deal with empirical data various types so I believe uh these two examples are very clever cleverly chosen to point some of the subtleties of modeling here cool section 97 models of false inference the section among other things contains a table of various settings that have been studied using active inference in the computational pathology space and the pathologies listed here with citations and a little bit of a note are addiction impulsivity and compulsivity delusions hallucinations interpersonal and personality disorders ocular motor syndromes pharmacotherapy and prefrontal syndromes visual neglect and disorders of interceptive inference under the section header of models of false inference

what would you say about this okay so I believe it links to recent developments and this new area of computational Psychiatry because unlike the traditional psychiatry in this new way of studying some abnormal behavior researchers try to develop computational models in order to model the normal human behavior and then study how quote-unquote false inferences can arise from those normal models and then try to map this kind of behavior onto various types of abnormal behavior abnormal psychiatric disorders so uh this can be a kind of introduction to this new research discipline and also uh I I'd like to uh point to Kristen's um review paper and I think it was published in nature a few months ago about computational Psychiatry so in that paper there are many more examples and other more many more details about how exactly those kind of computational models are constructed in order to study uh those disorders right and then and the interesting discussion we always have about what is the relationship between Bayes optimal generative models the ball rolling downhill things unfolding as they do in a setting and pathology and what does pathology mean what is an average what's a false belief or a hallucination in this setting and then 9.8 it's a very short summary they provide the six steps in figure 9.1 providing a generic method for Designing experiments and there's a lot of opportunities to ask different sorts of questions analogously to chapter five studying the mammalian nervous system here they highlighted their most active area of work which is on human computational pathology any other comments on chapter nine uh nothing particular comes to mind currently so